

Failure Analysis of a Titan PowrTwin 8900 Pump Cylinder

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ME-3033 Properties of Materials

May 01, 2026

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Overview

This failure analysis focuses on the fracture of a Titan PowrTwin 8900 pump cylinder (Part No. 144-832) and the cause of its failure. The Titan PowrTwin 8900 is an airless sprayer used for large-scale industrial and commercial painting; our client for this project, Ultimate Painting & Drywall, LLC, owned by Mr. Rufino Rios, primarily uses this machine for most of their jobs and currently has ten machines out-of-order due to the failure. Within this system, the cylinder encloses a pump mechanism and maintains the pressure required to deliver paint through the spraying system. A fracture of this component creates a leak through which pressure escapes, rendering the entire machine inoperable. We conducted both hardness tests and a microstructure analysis to identify the metal, which appears to be 440C stainless steel. The material characterization, fracture observations, and consultation with the equipment manufacturer and industry professionals show that the most probable cause of failure is hoop stress caused by residual water freezing and expanding inside the cylinder under cold conditions. Replacement of the damaged cylinder typically ranges from \$200–\$500, along with the costs of the equipment downtime, which can total up to \$10-15k per failure. We considered multiple solutions, including: insulation, heat tape, or heaters to prevent freezing; antifreeze additives to lower the freezing point; and removing the water through mechanical means such as compressed air or suction. Prioritizing ease-of-use in the solution, we implemented a valve-actuating pin in the inlet elbow to lift the ball valve and allow the trapped water to flow out through the inlet hose. This solution proved to be both effective in draining the water and well-liked by the operators in its simplicity and quick operation, and would cost around \$135 per retrofit on the side of the painting companies; if implemented in the manufacturing of the machine, it would likely add around \$65 per machine in manufacturing costs, not counting the cost of retooling and changing the manufacturing process.

Introduction

Airless paint sprayers are widely used in industrial and commercial painting applications to efficiently apply coatings to large surfaces. These systems rely on high-pressure pumping mechanisms that force paint through a spray assembly. A key component within this system is the pump cylinder (Part No. 144-832). The cylinder must be able to withstand cyclic pressure of 3300 psi as well as environmental exposure during normal service conditions. Failure of the pump cylinder prevents the sprayer from maintaining pressure, rendering the instrument inoperable. This leads to equipment downtime, which causes a chain of additional labor costs and delays in ongoing work that can regularly total \$10-15k in extra costs. Information obtained from Sherwin-Williams suggests that failures of this component are frequently observed, always in below-freezing temperatures. These failures are typically infant failures, often occurring within the first year of the pump cylinder's life, while the pump cylinder is expected to last five years of service at minimum. As the component is vital to the operation of the machine, understanding this failure will lead to improved reliability and can save tens of thousands of dollars per year per machine. Our client for this project was Ultimate Painting & Drywall, LLC, owned by Mr. Rufino Rios. Mr. Rios currently has ten Titan PowrTwin 8900 paint sprayers out of operation because of this failure.

Figures 1-4 provide an overview of the paint sprayer as well as a closer examination of the components within the system. Figure 1 shows the complete PowrTwin 8900, while Figure 2 presents an exploded view of the pump cylinder assembly, highlighting the individual components. Figure 3 presents a schematic illustrating the operation of the lower hydraulic system of the Titan PowrTwin 8900.

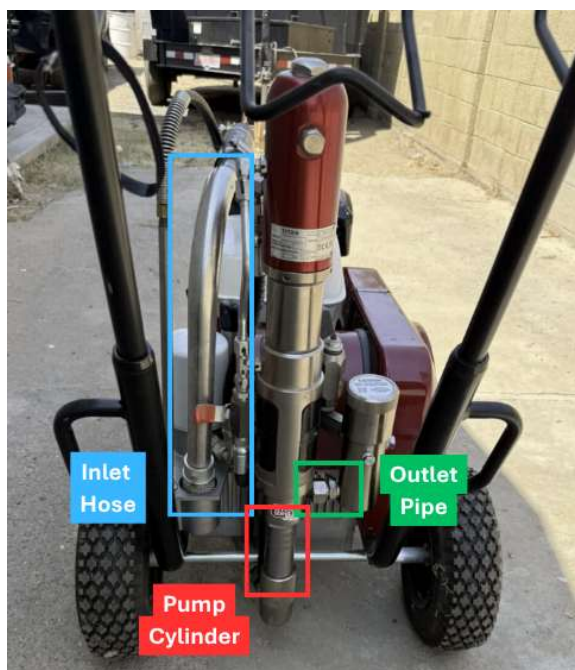


Figure 1: PowrTwin 8900

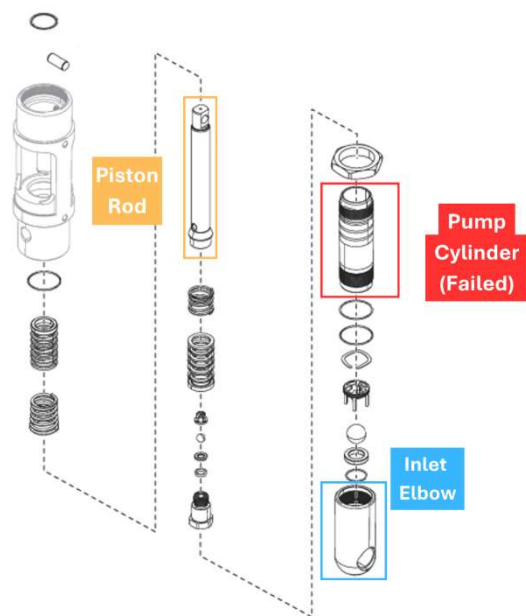


Figure 2: Fluid Section Exploded View

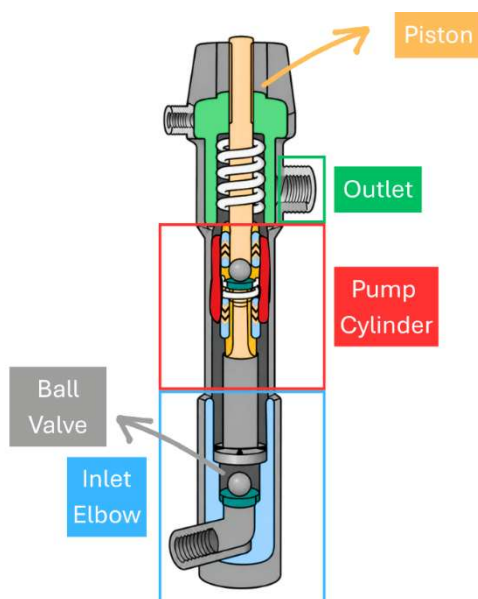


Figure 3: Schematic of Lower Hydraulic System and Ball Valve

Figures 4A and 4B show a failed pump cylinder in assembly, where the cylinder experienced a longitudinal crack along the length of the part.

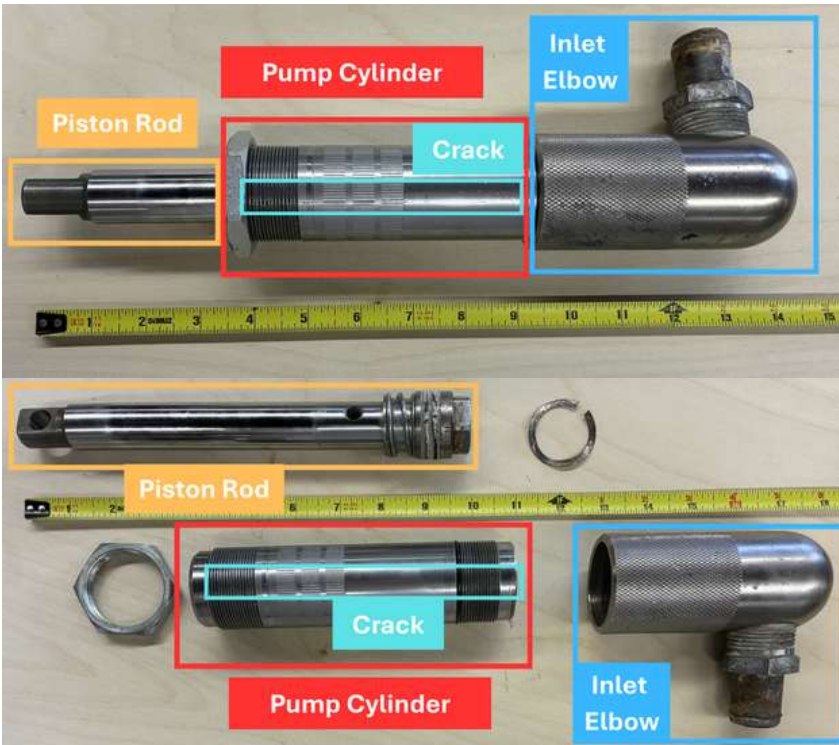


Figure 4A: Cracked Pump
Cylinder in Assembly

Figure 4B: Cracked Pump
Cylinder in Assembly –
Exploded View

Preliminary Examination

An initial visual examination of the failed pump cylinders shows very small amounts of wear on the inside surface and no evidence of corrosion. Pump cylinders failed both by cracking and leaking or by bursting, as seen in Figure 5. Besides the failure lines, no other cracks were found on the failed pump cylinders.



Figure 5: Two Different Failed Cylinders

A Rockwell C Hardness test was conducted on the cracked cylinder shown in Figure 5, yielding an average hardness of 58.7 ± 0.5 HRC. Figure 6 below shows the testing locations, while Table 1 shows the data gathered in step with the arrow direction noted on Figure 6.

Figure 6 (Left):
The Hardness Test
Locations



Table 1 (Right):
Hardness Test Data

	<i>HRC</i>
↓	58
	58.9
	59.1
	58.8
<i>Average</i>	58.7
<i>Std. Dev.</i>	0.48

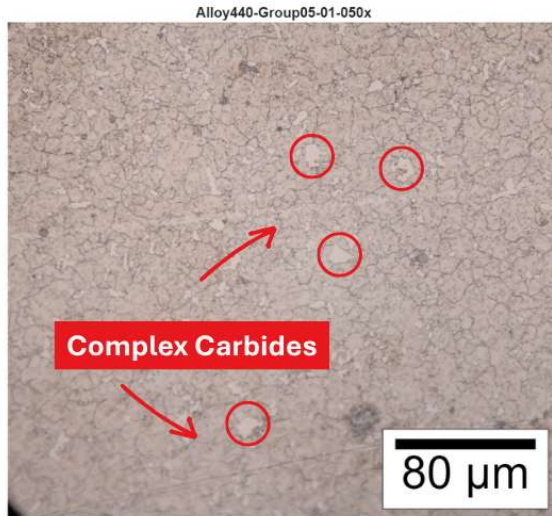
The Rockwell Hardness results suggested that the pump cylinders were made out of a high-grade martensitic stainless steel; this was confirmed in a call to the Titan Tool Technical Service line, where they stated that the pump cylinder was made from 440C stainless steel.

Microstructural Examination

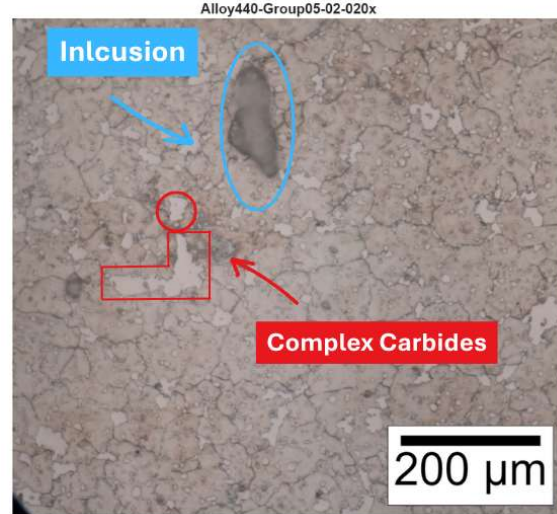


Figure 7: Microstructure Sample Piece from Burst Cylinder

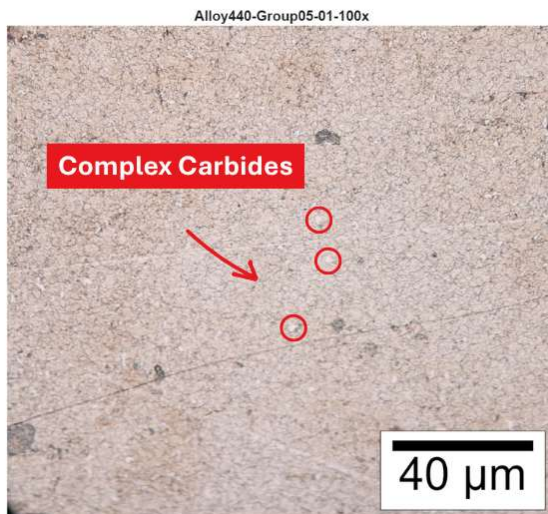
The microstructure sample was taken from the side piece of pipe that came from the burst pipe (see Figure 7 above). The sample was encased in epoxy, and the surface was wet ground to 600 grit and polished to a mirror finish. The first look through the microscope did not see any microstructure, so the sample was then etched with Villela's reagent to reveal the grain structure. In the microstructure pictures in Figure 8, the lighter areas are complex carbides, and the darker background areas either retained austenite or more likely tempered martensite.



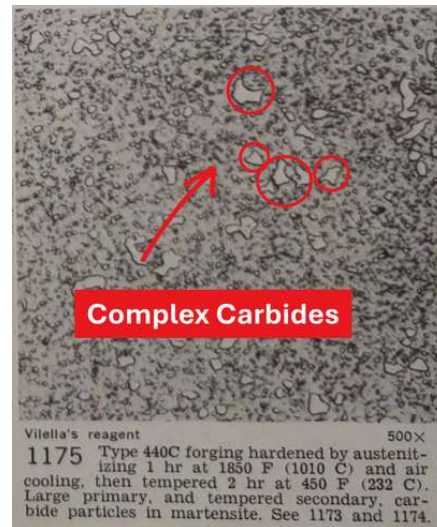
(A)



(B)



(C)



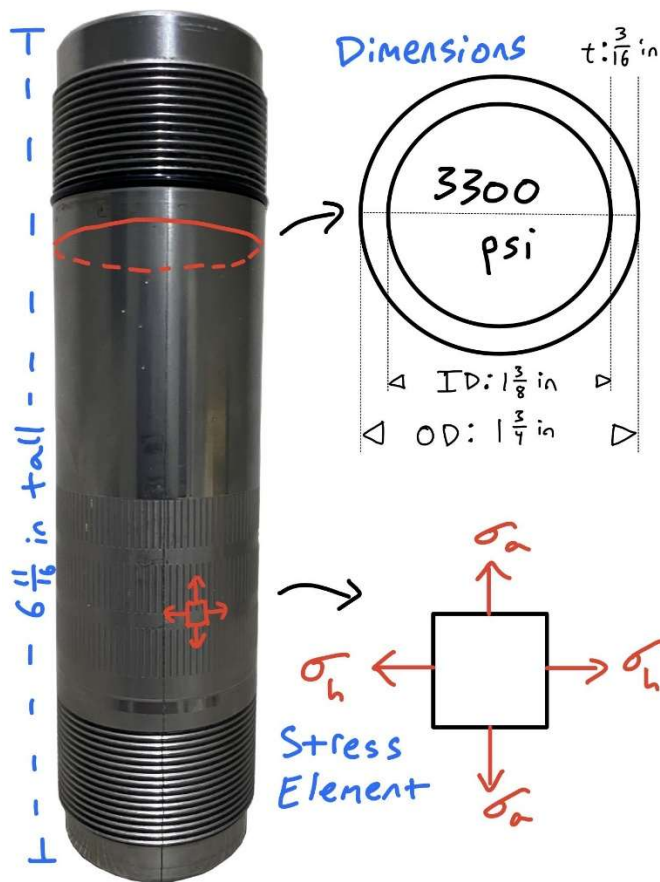
(D)

Figure 8: Microstructure at Various Resolutions and AMS Microstructure 1175

Of the various pictures provided in the AMS *Atlas of Microstructures of Industrial Alloys*, the sample microstructure photos best match pictures 1173-1179 (pp. 145-146), and principally 1175 (Figure 8D) which are of Type 440C stainless steel, as expected from the information given by Titan.

Stress Analysis

The pump cylinder is securely fixed in the lower fluid section of the pump assembly, and the main forces it is subject to are the cyclic axial and hoop stresses it undergoes while the pump sprayer is operating, which we assume to be uniform throughout the pump cylinder. Figure 9 displays a dimensioned cross-section of the pipe, as well as a representative stress element showing both the axial and hoop stresses.



<i>Inner Diameter</i>	$1\frac{3}{8}$
<i>Outer Diameter</i>	$1\frac{3}{4}$
<i>Wall Thickness</i>	$\frac{3}{16}$
<i>Length</i>	$6\frac{11}{16}$

Figure 9: Cross-Section and Stress Element

Table 2: Dimensions in Inches

Accounting for the thickness of the pump cylinder wall, the maximum expected interior hoop stress is just under 14 ksi, and the maximum axial stress is around 5.3 ksi, as described below.

Table 3: Calculation of Pump Cylinder Stresses

	Formula	Stress Value
<i>Axial Stress</i>	$\sigma_a = \frac{p_i r_i^2 - p_o r_o^2}{r_o^2 - r_i^2}$	5310 psi (35.4 MPa)
<i>Hoop Stress</i>	$\sigma_{h(max)} = \sigma_a + \frac{r_i^2 r_o^2 (p_i - p_o)}{r_i^2 (r_o^2 - r_i^2)}$	13930 psi (96.0 MPa)
	Where: p_i = internal pressure p_o = external pressure r_i = internal radius r_o = external radius	For: p_i = 3314.7 psia p_o = 14.7 psia r_i = 0.6875 in r_o = 0.875 in

Note that the stresses in Table 3 are still an approximation, as the given formulas assume closed ends, which are not directly true for the pump cylinder, though most likely the same forces are transmitted through its connection to the rest of the fluid section, as seen in Figure 10 below.

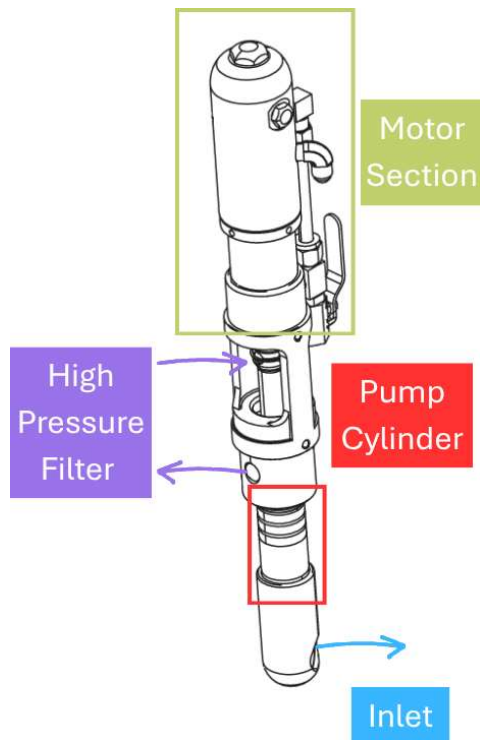
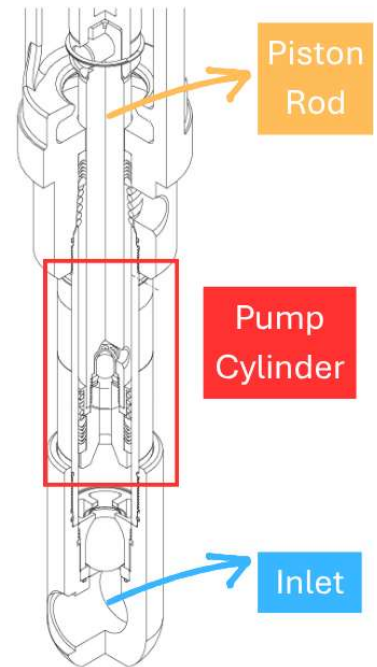


Figure 10
(Left):
Full Fluid
Section
Assembly

Figure 11
(Right):
Cutaway of
Lower
Fluid
Section



Discussion

The PowrTwin pump cylinder (Part No. 144-832) is designed for cyclic pressurization as part of the pump and fluid system of the Titan 8900. The system's inlet hose in the Titan PowrTwin 8900 provides paint into the chamber created by the pump enclosed by the cylinder, where paint is pressurized and released through the outlet hose. The pump cylinder operates in a reliable manner under typical conditions but is susceptible to fracture when temperatures reach below the freezing point of water. These fractures do not occur while in use, only once the machine is stored in cold conditions; the pump cylinder is stored full of water, as at the end of the day the machine is flushed with water to clean out residual paint. When maintained and stored in ideal conditions, the component is able to withstand a maximum operating pressure of 3300 psi and has a lifetime of about 5 years. The component we analyzed was in use for less than a year and was not used on a daily basis. The presence of a crack in the cylinder wall is an indicator of a failure due to unexpected conditions, not an end-of-life failure due to wear. The unexpected early fracture along with tests suggests the main factor of the failure was due to abnormal stress from the force of the expanding water as the water froze.

To analyze the pump cylinder, we first decided to identify the material of which it was composed. Hardness testing and microstructural analysis were performed, where we confirmed our metal is most likely 440C stainless steel, as specified by the manufacturer of the cylinder. The hardness tests showed the metal to have an average hardness of 58.7 HRC. The microstructure allowed us to compare its characteristics to those of 440C stainless and found them to match closely. These steels are frequently heat-treated to improve strength, hardness, and wear resistance, but the increased hardness can also lead to an increase in brittleness. The material may be more susceptible

to cracking under extreme stress conditions. No obvious material defects were observed that would be indicators of a manufacturing flaw. According to Sherwin-Williams, a paint-supply store that also fixes inoperable paint machines, fractures in this specific component are common during the winter when the temperature reaches below the freezing point of water. Mr. Rios has ten of these PowrTwins out of use due to this same failure. Many have a minute crack, invisible until the sprayer is turned on and the cylinder leaks; a few have fully burst with pieces broken and falling apart. Although it is difficult to see in the image, a small leak is visible and highlighted for clarity in Figure 12.



Figure 12: Leaking Pipe – Only Visible While Running

Consultation with the equipment manufacturer Titan Tool, Sherwin-Williams representatives, and Mr. Rufino Rios suggested that the residual water inside the cylinder, trapped by the ball valve, may have contributed to the fracture. Every case where a pump cylinder failed was in below-freezing conditions, often overnight. At the end of the workday, the paint sprayers are rinsed

through with water and stored in the work van. While Ultimate Painting & Drywall has a policy of not operating the paint sprayers in conditions below 40°F, many locations often dip below freezing overnight, and the remaining water filling the pump cylinder—stopped on the bottom by the ball valve—freezes, which puts large stresses on the cylinder while it is stored overnight, as the volume of water expands by 9% upon freezing. Figure 13 shows that even -10°C produces more stress in the cylinder than it is intended for in normal operation, 112 MPa compared to the 96 MPa calculated in Table 3.

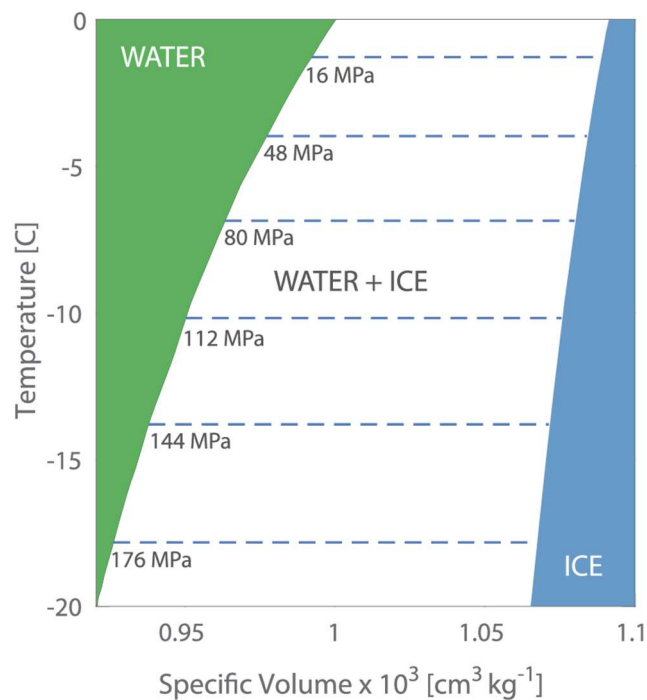


Figure 13: Stresses of Water Freezing in a Constant-Volume Container

A Titan Tool winterization flyer (included in Appendix B) recognizes that water in the sprayer can “damage the sprayer’s fluid section” by “driving the internal pressures to 10,000 psi,” which corresponds to 42,250 psi (291 MPa) of hoop stress following the methods of Table 3.

Economic Analysis

The costs of the pump cylinder failing result in material costs, repair costs, and operational downtime. Purchasing new pump cylinders for the Titan PowrTwin 8900 costs \$200-\$500 (4-12% of the total machine cost) each, depending on the supplier. Moreover, installation of the new cylinders is labor-intensive and time-consuming. For many painting jobs around the continental US, Ultimate Painting & Drywall dispatches only one Titan PowrTwin 8900, and its failure results in at least two weeks of downtime, on average, which costs at least \$3-4k in labor per week. Addressing a failed cylinder often comes down to two approaches: (1) waiting for the paint sprayer to be repaired at the nearest Sherwin-Williams, which can take over a month, or (2) seeking out local painters and painting companies and trying to rent another paint sprayer to begin work. Both options significantly eat into the profits made from a painting job. Overall, this failure of the pump cylinder can regularly cost Ultimate Painting & Drywall around \$10-15k in repair and downtime.

Current and Possible Solutions

The painters of Ultimate Painting & Drywall already use two practices to address this failure; however, the existing solutions are cumbersome and often not realistic or workable. The first is wrapping the paint sprayer in insulation at the end of the workday; this does reduce the rate of failure but is inconsistently done as it requires at least half an hour of extra work by the operators, which, at the end of a long day of work, is a difficult proposition. The second is carrying a heater in the work van; however, these heaters are difficult to transport in the often-stuffed work van, and they require access to electrical outlets to power the heaters, which are often restricted (such as in hotels) due to fire safety concerns. These current solutions address the failure by attempting to limit the heat transfer and cooling of the water-filled cylinder overnight. Alternatively, the problem

can be addressed by lowering the freezing point of the rinsing fluid or by removing the water from the cylinder by mechanical means.

The Titan Tool winterization flyer (Appendix B) recommends running mineral spirits or Titan LS-10 Liquid Shield™ Plus fluid, rather than just water, through the machine, which would cost \$15-20 per rinse. Adding antifreeze additives such as tetrahydrofuran, 1,3-dioxylene, or MTHF to the rinsing water may achieve the same objectives while costing \$10 or less per rinse. However, non-recommended antifreeze carries the possibility of damaging internal components or seals in the machine. Moreover, either method would require a plan to dispose of the replacement fluid, as these chemicals cannot be simply drained out on the ground. These methods would not require modification of the paint sprayer, but would incur recurring costs and add complexity to the end-of-day process for the operators, which is likely to reduce process adoption.

On the other hand, mechanical methods of removing the water would be more difficult to implement initially but could be both cheaper and easier to use in the long run. Three main methods of removing the water were considered: blowing the water out by running compressed air through the system; sucking out the water with a suction device that attaches to the outlet paint pistol; or actuating the ball valve to allow the water to flow out through the inlet hose. Using compressed air would be quicker for the operators than antifreeze but would require both significant modification of the inlet hose to create a simple connection and the addition of a compressed air tank to the work van, which would take up valuable space. A suction device may be more maneuverable and space-efficient but would be both very difficult and more expensive to design and manufacture. Using gravity to remove the water by actuating the valve is the simplest of the

mechanical solutions; we considered laying the machine on its side to allow the water to drain, but that would cause the lubrication at the top of the piston to drain as well, which is not acceptable. Instead, a valve-actuating pin, inspired by such a pin on the Titan Impact 440 paint sprayer, a smaller model for which the pin is used to break up dried paint blocking the ball valve, could allow the operators to actuate the valve and drain the water through the inlet hose, adding only 30 seconds to the end-of-day process. While this approach does require modification of the machine, it requires less design and investment than the other options and offers the easiest operation for the paint sprayer operators. Thus, we chose to implement a valve-actuating pin retrofit, discussed below in the Implementation section.

Conclusions

1. The cylinder encloses the pump mechanism to create a chamber where paint is pressurized.
2. The pump cylinder from the Titan PowrTwin 8900 is composed of 440C stainless steel.
3. The principal cause of fracture is residual rinse water reaching its freezing point and expanding within the cylinder overnight, creating large internal pressures (≥ 112 MPa) in excess of operational pressures (96 MPa).
4. Each fracture is estimated to cost Ultimate Painting & Drywall around \$10-15k, on average, through repair and downtime costs.
5. Current methods of addressing this failure are labor-intensive and inadequate; freezing temperatures at night cannot be avoided, and so a workable solution must be found to prevent future failures.

Recommendations

1. To prevent freezing, heat loss from the cylinder could be reduced by heat tape, heating, or insulation. Heat tape is relatively inexpensive (around \$5 at the lower end), but would be a recurring seasonal cost and require a power source; van-mounted heaters would cost around \$80 and also require addressing power-supply. Insulating the work vans would cost around \$800, but could synergize with heating to protect the residual water from freezing.
2. Alternatively, the rinsing fluid could be modified with antifreeze such as THF, 1,3-dioxylene, or MTHF. This would cost around \$10 per rinse (essentially \$10 per day of operation) and add complexity to the end-of-day process, making it less likely to be performed consistently. Disposal and compatibility with sprayer components would also have to be addressed.
3. Mechanical removal of water through compressed air, suction, or gravity could also prevent freezing stresses. Compressed air would cost around \$120 to modify the machine and add a tank to the van. A suction device would be easier for an operator to use, but would be much more difficult and costly to design and manufacture. Adding a pin to actuate the ball valve would be simple for the operators to use daily and would require modification of the main body of the machine at a cost of \$135.

Summary Table of Recommendations

Recommendation	Advantages	Disadvantages
Heat Tape	• Low upfront cost • Simple concept	• Requires reliable power source • Recurring seasonal cost • Requires activation each night
Heater in Van	• No modification to machine required • Can maintain overall temperature of equipment	• Bulky and difficult to transport • Requires external power • Not practical for travel
Insulation	• No power required • Passive protection	• Requires significant modification of the work van • May not prevent failure in extremely cold environments
Antifreeze	• Lowers freezing point of rinsing fluid • Reduces risk of expansion and cracking	• Recurring cost • Hazardous waste • Potential compatibility issues with seals and components • Requires extra operator effort
Compressed Air from Inlet	• Reusable system	• Requires air tank • Takes up space in van • Needs refilling
Outlet Suction	• Easy to use by operator • Does not require modification of existing parts	• Difficult to design and implement • Requires power supply
Valve-Actuating Pin	• Extremely easy for operators to use • Integrated into sprayer • No recurring costs • Reliable even in very cold situations	• Requires design modification • Large initial implementation effort

Implementation of Valve-Actuating Pin

Considering the advantages and disadvantages listed in the preceding table, we discussed possible solutions with Mr. Rios. He emphasized ease of operation to improve the chance the operators would use the solutions at the end of the workday, and recommended the valve-actuating pin over our other recommendations. Thus, we decided to implement a valve-actuating pin and evaluate if it can effectively address the failure of the pump cylinder. This idea was inspired by the pin component in the Titan Impact 440, a smaller paint sprayer in which the pin serves to break up dry paint that can block the ball valve. Figure 14 shows the pin assembly for the Impact 440.

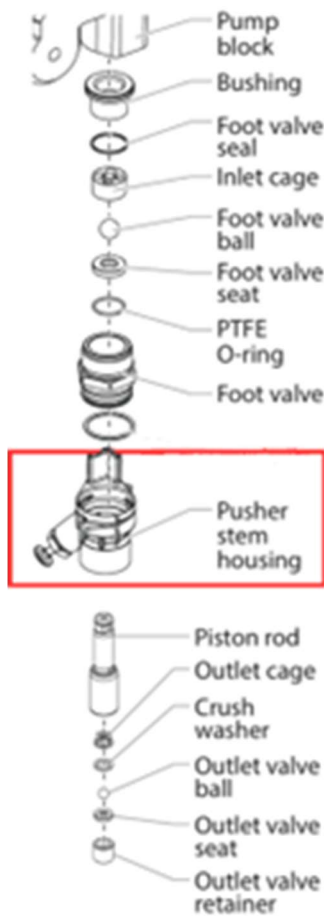


Figure 14A: Impact 440 Fluid Section Exploded



Figure 14B: Impact 440 Pusher Pin Assembly



Figure 14C: Impact 440 Pusher Pin Assembly Exploded

Our new design is composed of a valve-actuating pin assembly concentric with the inlet elbow (and pump cylinder). The pin assembly consists of a 2-part press-fit pin, a 3D-printed plastic cap, a welded, threaded pipe, and a spring. When pressed, the pin raises the ball valve and allows the trapped fluid to flow out through the inlet hose. Table 4 shows the expected costs for the retrofit of an existing Titan PowrTwin 8900 Plus at \$135. Figure 15 shows the implemented assembly.

Table 4: Valve-Actuating Pin Cost Breakdown

Item	Cost
Inlet Elbow	\$0 – Retrofit Existing Elbow
Materials	
- Brass and steel round bar stock for pin	
- Pipe for housing	\$15
- Spring	
- 3D-printing the cap	
Machine Time and Welding	\$120
Total	\$135

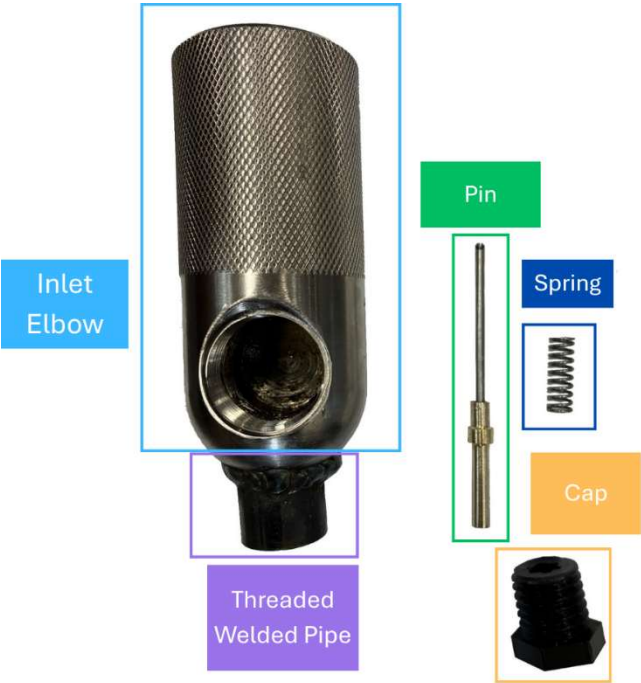


Figure 15A: Valve-Actuating Pin Assembly Exploded



Figure 15B: Valve-Actuating Pin Assembly

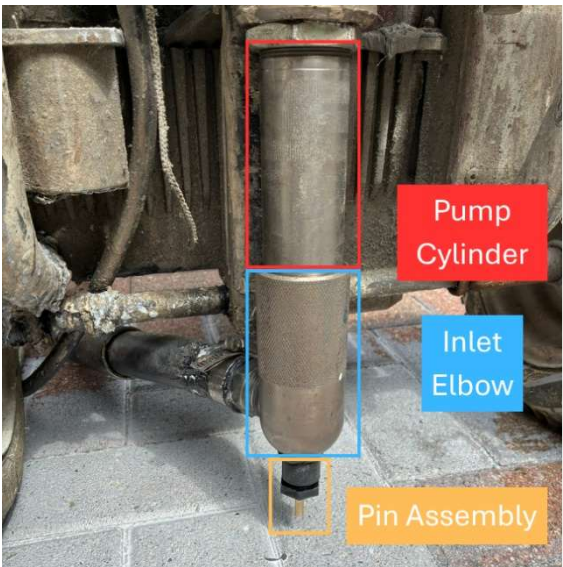


Figure 15C: Pin Assembly Installed

When tested on a regularly-used PowrTwin 8900 Plus (Figure 15C), the valve-actuating pin assembly proved to not affect the operation of the paint sprayer and successfully evacuated the water from the pump cylinder upon shutdown. Using the pin at the end of the test took Mr. Roberto, a painter who has been working with Ultimate Painting & Drywall for three years, about 20 seconds. He expressed that this method was much quicker than the various methods currently used to prevent water freezing in the pump cylinder, and felt that this modification would simplify and streamline his cleaning process at the end of a workday. Thus, this solution proves to be practical, implementable, easy-to-use, and relatively cheap. If this modification can prevent even one failure, with \$135 of investment and 20 seconds of operation, it will have paid for itself 80 times over.

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Checklist

Failure Analysis Final Report Checklist

Please fill out and include this checklist as the last page on your report.

Report Title: _____

Student authors: _____

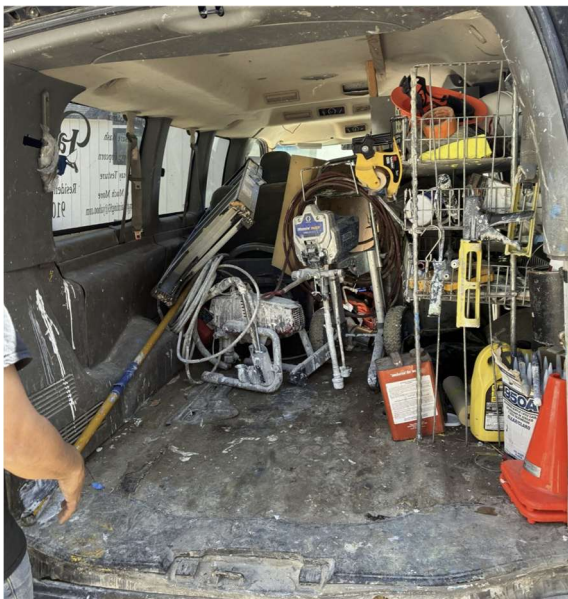
✓	Check boxes in left column to show that you have verified each item in your report
✓	The report is submitted as a pdf by the due date and time. The filename includes the type of component and the author's names. (Ex: Tacoma+Hinge+Pin+Failure+Analysis-Bickle,+Denton,+Holman.pdf) No Word doc submittals!
✓	Title Page includes a title <i>specific to your project</i> , the date, and the names of the authors.
✓	All pages are numbered.
✓	The line spacing in this report is DOUBLE SPACED .
✓	There is a table of contents with page numbers.
✓	Headings for the major sections of the report (Introduction, Results, etc.) are included.
✓	The <i>Overview and Summary of Conclusions and Recommendations</i> section summarizes your answers to the four main questions: How does it work? What is it made of and how was it made? Why did it fail? How can we improve it?
✓	Each figure is numbered and includes a descriptive caption.
✓	A stress analysis section with free body diagram showing all forces is included.
~	All photos include a scale or magnification. Some part-in-environment photos do not
✓	The Conclusions are numbered. They are supported in the Discussion section.
✓	The Recommendations are numbered and specific. They are supported in the Discussion section. Sectioned by manner of preventing failure
✓	We have compared our report to the failure analysis section of the textbook to ensure all important sections, results, etc., are included.
✓	To the best of our ability, we have edited this report for conciseness, style, grammar, spelling, etc.

Appendix A: Photo Gallery



Titan PowrTwin 8900 without the modifications. Seen at one of the job sites Ultimate Painting & Drywall, LLC on April 8th, 2026.

Figure A1: A working Titan PowrTwin 8900



The work van of our client, Mr. Rufino Rios. Space is at a premium inside these work vans. Also taken on April 8th, 2026.

Figure A2: Interior of Work Van

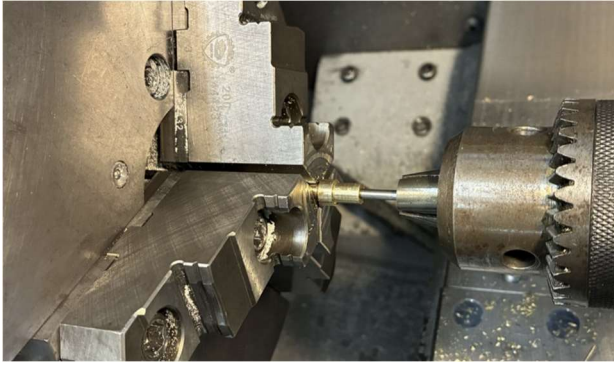


Figure A3: Press-fitting of new pin

Press-fitting of the two parts of the valve-actuating pin. Taken April 29th, 2026 in the McElroy Prototyping Lab [Machine Shop].



Figure A4: Non-modified inlet elbow full of water

This is the non-modified inlet elbow of the Titan 8900 PowrTwin we used to test our valve-actuating pin solution. The trapped water is easily visible when the elbow is removed. This water causes the connected pump cylinder to rupture in freezing conditions. Taken April 30th, 2026, in testing the valve-actuating pin on a working machine.



Figure A5: Installed valve-actuating pin on Titan 8900

Titan PowrTwin 8900 with the integrated valve-actuating pin on the inlet elbow. While the valve-actuating pin is closer to the ground than the unmodified elbow, there were no clearance issues in testing, nor are there expected to be in normal operation. Also taken April 30th, 2026

Appendix B: Titan Tool Winterization Flyer



TITAN SERVICE REMINDER

WINTERIZE YOUR AIRLESS SPRAYER

As we approach the winter season and the inevitable cooler temperatures, it is important to remember to protect your Titan sprayer from the cold. Below are some tips to make sure your sprayer is ready to work when you are.

WINTERIZING 101:

The best protection against freezing is to run Mineral Spirits through the system after you have finished cleaning.

Mineral Spirits will not freeze. It also protects against corrosion and keeps the paint pump lubricated. If a sprayer freezes with water or a water-based coating, it can drive the internal pressures to 10,000 PSI. This may damage the sprayer's fluid section or pressure control components.

ADDITIONAL TIPS:

- Never store the sprayer under pressure. This may cause over-pressurization and is a safety hazard. Leave the prime/spray or bypass valve in the priming position.
- When transporting the sprayer between job sites or leaving it overnight in a vehicle, do not leave the sprayer full of water or water-based coatings.
- Store the sprayer in a heated area between jobs.
- If spraying water-based coatings and storing with mineral spirits is not practical, run Titan LS-10 Liquid Shield™ Plus undiluted through the sprayer and relieve the pressure.
- If the fluid section does freeze, allow it to thaw before running the sprayer to avoid damaging the sprayer.
- **Sprayers with a gasoline engine:** See the engine manufacturers' manual for complete winterization instructions; however, it is generally recommended to drain all gas from the engine, followed by running the engine until it stops, prior to long-term or seasonal storage. This should be done with the sprayer in priming position and the pressure set to minimum.



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